Title: Identifying Conditions Favouring Multiplicative Heterogeneity Models in Network Meta-Analysis

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Configurations:

R version 4.5.1 (2025-06-13)

Platform: aarch64-apple-darwin20

Running under: macOS Tahoe 26.3.1

Matrix products: default

BLAS:
`/System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib`

LAPACK:
`/Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib`; LAPACK version 3.12.1

locale: [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Toronto tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] xtable_1.8-4 patchwork_1.3.2 ggpattern_1.3.1 gflnma_0.0.0.9000 [5] ggplot2_4.0.3 tibble_3.3.0 dplyr_1.1.4 netmeta_3.4-0
[9] meta_8.3-0 metadat_1.4-0 metabook_0.2-0

loaded via a namespace (and not attached):

[1] magic_1.6-1 generics_0.1.4 xml2_1.4.0 stringi_1.8.7
[5] lattice_0.22-7 lme4_1.1-37 hms_1.1.3 digest_0.6.37
[9] magrittr_2.0.4 grid_4.5.1 RColorBrewer_1.1-3 mvtnorm_1.3-3
[13] CompQuadForm_1.4.4 Matrix_1.7-3 purrr_1.1.0 scales_1.4.0
[17] numDeriv_2016.8-1.1 abind_1.4-8 reformulas_0.4.1 Rdpack_2.6.4
[21] cli_3.6.5 rlang_1.1.6 rbibutils_2.3 splines_4.5.1
[25] withr_3.0.2 tools_4.5.1 tzdb_0.5.0 nloptr_2.2.1
[29] minqa_1.2.8 metafor_4.8-0 colorspace_2.1-1 mathjaxr_1.8-0
[33] boot_1.3-31 vctrs_0.6.5 R6_2.6.1 lifecycle_1.0.4
[37] stringr_1.5.2 MASS_7.3-65 pkgconfig_2.0.3 pillar_1.11.1
[41] gtable_0.3.6 glue_1.8.0 Rcpp_1.1.0 tidyselect_1.2.1
[45] rstudioapi_0.17.1 farver_2.1.2 nlme_3.1-168 igraph_2.1.4
[49] readr_2.1.5 compiler_4.5.1 S7_0.2.0

Code execution process:

Before running any scripts, set the folder `Code Supplement` as the working directory.

The following scripts can be executed independently and do not require a specific order:

- `compare_AICs.R`
  Reproduces the histogram in Figures 1, and Appendix A.3 Figure 10, Table 1, Table 2

- `run_four_case_studies`.R‘
  Reproduces the results for Case Studies 1–4, including Figures 2–8 and Appendix A.6 Tables 3–6.
- `run_appendix_case_studt.R`
  Reproduces the Appendix A.4 case study illustrating the correct calculation of standard errors.

Source of data:

The `nmadb` package does not contain the datasets directly. Instead, it provides functions for retrieving data from an external REDCap database hosted at the Institute of Social and Preventive Medicine (ISPM), University of Bern, as described in the package documentation.

The package source code is archived on CRAN:

https://cran.r-project.org/src/contrib/Archive/nmadb/?C=D;O=A

Since the `nmadb` package is no longer actively maintained and relies on online access to the external database, we downloaded and locally saved all datasets used in this paper to ensure long-term reproducibility. The script `savedata.R` contains the code used to retrieve and store these datasets locally.

The datasets are stored in `nmadb_twoarm_data_all.rds`